

SIMATS ENGINEERING

ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE:** CSA1304
COURSE OUTCOME COVERED CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks: 25

Duration: 1 Hour

Simulation

Code of Conduct:

I S Venkata Abhishek - 192311053 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S Venkata Abhishek

1. Design the regular expression for the language represented by the given automata

Given

From the automaton:

- $q_0 \rightarrow q_1$ on **0 or 1**
- $q_1 \leftrightarrow q_2$ on **0 or 1**
- q_2 is the **final state**

The machine accepts all binary strings having **even length greater than or equal to**

Regular Expression

$((0+1)(0+1))^+((0+1)(0+1))^*((0+1)(0+1))^+$

or equivalently,

$(0+1)(0+1)((0+1)(0+1))^* (0+1)(0+1)\big((0+1)(0+1)\big)^*(0+1)(0+1)((0+1)(0+1))^*$

Language:

All binary strings over $\Sigma=\{0,1\}$ having **even length (2,4,6,...)**

2. Consider the following grammar $S \rightarrow aSc | T | U | \epsilon$ $T \rightarrow bTc | \epsilon$ $U \rightarrow aUb | \epsilon$ A) What is the language defined by the grammar? B) Give the derivation tree for the word aaabbc C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify

Given Grammar

$S \rightarrow aSc \mid T \mid U \mid \epsilon$

$T \rightarrow bTc \mid \epsilon$

$U \rightarrow aUb \mid \epsilon$

A) The grammar generates the union of the following languages:

- $S \rightarrow aSc$ generates strings with equal numbers of **a's** at the beginning and **c's** at the end.
- $T \rightarrow bTc$ generates strings of the form:

$L(T) = \{b^n c^n \mid n \geq 0\}$

- $U \rightarrow aUb$ generates strings of the form:

$L(U) = \{a^n b^n \mid n \geq 0\}$

Hence, the language generated by the grammar is

$$L = \{anxcn \mid x \in L(T) \cup L(U) \cup \{\epsilon\}, n \geq 0\}$$

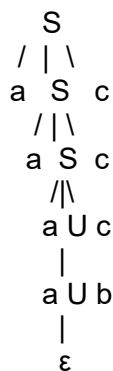
That is, the grammar generates strings with matching numbers of **a's** and **c's**, while the middle part may belong to b^nc^n and a^nb^n , or be empty.

B) Derivation Tree for aaabbc

Leftmost derivation:

$S \Rightarrow aSc$
 $\Rightarrow aaScc$
 $\Rightarrow aaaUbcc$
 $\Rightarrow aaa\epsilon bcc$
 $\Rightarrow aaabbc$

Derivation Tree



C) Yes, the grammar is ambiguous.

The string ϵ can be derived in more than one way.

First derivation: $S \Rightarrow \epsilon$

Second derivation: $S \Rightarrow T \Rightarrow \epsilon$

Third derivation: $S \Rightarrow U \Rightarrow \epsilon$

3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression 00^*1+11^*0

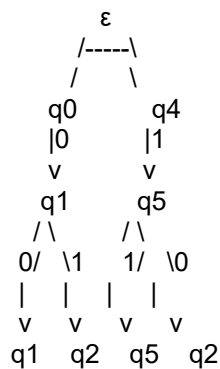
Regular Expression

$00^* 1+11^* 000^* 1+11^* 000^* 1+11^* 0$

This expression represents the union of:

- $00^* 100^* 100^* 1$
- $11^* 011^* 011^* 0$

ϵ -NFA



q2 = Final State

Transition Table

State	0	1	ϵ
$\rightarrow q0$	q1	—	q4
q1	q1	q2	—
q4	—	q5	—
q5	q2	q5	—
*q2	—	—	—

4. I) Show that if $L1$ and $L2$ are regular then $(L1+L2)$ also regular li) Is the language $\{ a^p \mid p \text{ is prime} \}$ regular? Justify

Assume $L1$ And $L2$ are regular languages.

Then there exist DFAs

$M1=(Q1,\Sigma,\delta1,q1,F1)$ and $M2=(Q2,\Sigma,\delta2,q2,F2)$

that accept $L1$ and $L2$.

Construct a new automaton that accepts the **union** of both languages.

Since finite automata are closed under **union**, the new automaton also recognizes a regular language.

Therefore,

$L_1 \cup L_2$ (or $L_1 + L_2$) is **regular**.

Hence Proved.

(ii). Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify.

The language

$L = \{a^p \mid p \text{ is prime}\}$

Proof using Pumping Lemma

Assume that L is regular.

Let the pumping length be m .

Choose $w = a^p$

where p is a prime number and $p > m$

According to the Pumping Lemma,

$w = xyz$

such that

- $|xy| \leq m$
- $|y| > 0$

Let

$y = a^k, k > 0$

Pump y twice.

$xy^2z = a^{p+k}$

The length becomes $p+k$

which is **not necessarily prime**.

Hence,

$xy^2z \notin L$,
 $\{a^p \mid p \text{ is prime}\}$ is NOT Regular.

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata

equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states

1: Initial Partition

Final states

$$F = \{q_1, q_2, q_4\}$$

Non-final states

$$Q - F = \{q_0, q_3, q_5\}$$

2: Equivalent States

After partition refinement,

Equivalent states are

$$q_1 \equiv q_4$$

No other states are equivalent.

Hence,

Merge

$$(q_1, q_4)$$

3: Minimized States

The minimized DFA has

- A = q_0
- B = q_3
- C = (q_1, q_4)
- D = q_2
- E = q_5

4: Transition Table

State	0	1
$\rightarrow A(q_0)$	B	C
B(q_3)	A	C
*C(q_1, q_4)	D	E
*D(q_2)	D	E
E(q_5)	E	E

Initial State q_0

Final States (q1,q4), q2